PART 8

Consist system

Trains made of real vehicles: built from the library, spaced by their couplers, driven from any car, split and joined on the timeline, and pushed backwards into sidings without any special cases.

8.1 What the prototype does

8.2 Target consist model

8.3 Spacing maths

8.4 Coupling and uncoupling

8.5 Driving from any car

8.6 Train Builder UI

8.1 What the prototype does

CURRENT STATE (LATEST SOURCE)

- **Consist** box: wagon list with **+/−**. Each wagon is a `copy()` of the root with its own Follow Path. Its offset is the gap to the car ahead (default copied from the previous wagon).
- **Wagon Data**: a `vehicle_root` field stub (“library not built yet, always clones the root”).
- The root keeps a **cumulative route distance** and a **history log** of curve changes (`distance, from/to curve, offsets, direction`). Junctions are solved geometrically so spacing doesn't drift by a metre at each switch.
- A trailing wagon that reaches a fork first (for example when reversing) decides it itself and writes it to the shared log.
- **Fix Train** reruns the sync three times. It also runs automatically when the Drag Slider goes idle. **Consist Debug** lists the log.

This is the right *idea* (distance-based, shared history, whichever end leads decides). Part 5's occupied path is the same idea with the history replaced by turnout states, which removes the dedupe/prune machinery. The prototype is a good test bed for checking the new evaluator: both should put every wagon in the same place on the same test scenes.

8.2 Target consist model



Every member is a full vehicle (own root, pivots, rig). The tender is its own vehicle, coupled with gap ≈ 0 by a drawbar.

Figure 8.1 A consist is an ordered list of vehicle instances plus one distance channel.

8.3 Spacing maths

With each vehicle's root at its centre and coupler distances in its definition (Part 6 §6.3), the reference distance of member *k* behind the anchor is:

```
d(0) = consist.distance
d(k) = d(k−1) − [ rear_coupler(k−1) + gap(k) + front_coupler(k) ]       (facing −1 swaps front/rear)

pivot point of member k:   d(k) + facing(k) × pivot.offset
```

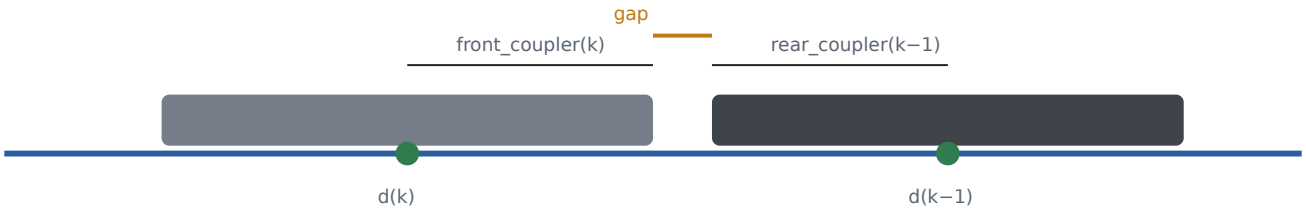


Figure 8.2 Spacing is measured **along the route**, so cars stay correctly spaced through curves and switches.

8.4 Coupling and uncoupling

From session 5: “If the user uncouples, the cars stay on the curve but don't react to dragging… maybe a decouple mode lets the drag slider on a car's own root work on its own.” As an animatable design:

Operation	What the add-on does
Uncouple at frame F (between member k−1 and k)	Creates consist *B* from members k… Its distance channel is keyed at F to the exact value that keeps each car where it is, then held flat (the cars stand still). Consist A keeps members 0…k−1. From F onwards the two move independently. Before F, the pipeline still treats them as one train (a “split event” stored on A: `split_frame`, `child_consist`).
Couple at frame F	Use **Arrive At** to bring the couplers together, then Couple. From F, B's members are appended to A with the measured gap, and B's distance channel is ignored after F.
Drag an uncoupled car	Selecting any car of B drives B. It has its own Drag Slider target, which is exactly the “decouple mode” idea.
Run-round / shunting	Just sequences of the above, with turnout keys (Part 5). The loco uncouples, runs round via the loop, and couples to the other end: facing is unchanged, but it becomes member 0 of a re-ordered list.

DECISION

Should split and join be **timeline events** (so scrubbing before F shows the joined train) or **destructive edits** (make two consists and animate each)? Recommendation: **timeline events**. Shunting scenes are the main reason to build consists, and they need to scrub correctly.

8.5 Driving from any car

- The Drive box acts on **the consist of the selected object**, whether that is a loco, wagon, rig bone or mesh. The add-on walks up from any selected part to its vehicle, then its consist.
- Pushing (the loco at the back) needs nothing special: when distance decreases, the leading end is the rear member (Part 5).
- **No collisions.** As agreed in session 5, running a train into itself or another train just clips. An optional overlay can flag overlap between consists on the same path segment in red, as a hint rather than a barrier.

8.6 Train Builder UI

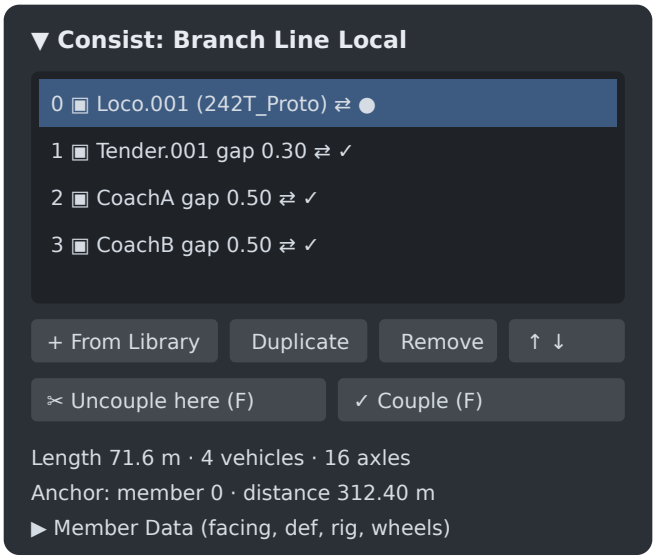


Figure 8.3 Mock-up of the finished Consist box (not built).

⇄ flips a vehicle's facing (runs backwards)

✓ = coupled to the car ahead (keyed state)

“From Library” opens the Asset Browser, filtered to Track It vehicles (Part 9)

Row order = train order; drag to reorder

The prototype's clone-the-root behaviour stays as “Duplicate” for quick proxies

Members are linked library overrides or local copies, so ten identical coaches share one mesh and rig.